Bahul Neel Upadhyaya

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Professional Summary

Technology leader with extensive experience in software architecture and engineering, focusing on enterprise-scale solutions that improve business operations. Combines technical expertise with business understanding to deliver practical, effective solutions. Experienced in building and leading technical teams. Advocates for workplace neurodiversity, bringing different perspectives (ADHD diagnosed and pending ASD assessment) to problem-solving and innovation.

Professional Experience

Undisclosed Client (under NDA)

Contract Developer/Architect *June 2024 - May 2026*

Architected and delivered an event-planning platform from concept to launch-ready over a two-year engagement. The project is complete but unlaunched at the close of the engagement; product, client, and implementation details remain under NDA. The work served as the production crucible in which several of the patterns now published as open research and libraries under RelationalFabric — Howard's claims engine, Canon's lazy typing, and Suss / RaCSTS propagator networks — were tested, refined, and proved out at the scale of a real codebase.

- Delivered the platform end-to-end using AI-assisted development practices, sustaining a high delivery velocity across the engagement
- Underpinned the architecture with a rules-driven platform model built on a homegrown rules engine
- Implemented a data architecture on FlureeDB for FAIR data storage with native JSON-LD support
- Executed an offline- and mobile-first product strategy across web and mobile surfaces
- Stood up cloud infrastructure on AWS sized for projected scalability requirements
- Acted as the proving ground for the public RelationalFabric research track now visible in the Projects and Writing sections of this site

Hybr Ltd

Senior Full-stack Web Developer *June 2023 - May 2024*

Led technical improvements and platform modernization, including semantic web implementation and design system development.

- Developed comprehensive design system to improve development consistency
- Introduced semantic web architecture and knowledge engineering concepts to enhance internal modelling and data integration
- Improved platform performance and user experience through targeted updates

Contxt

Contract ML Architect and Senior Developer *August 2022 - February 2023*

Guided architectural evolution for API classification systems, combining ML solutions with platform modernization.

- Guided team's architectural evolution from monolithic to pluggable multi-layer design supporting cross-platform deployment
- Developed synthetic data generation system for classifier testing
- Investigated language model applications for API content classification

Elsevier (RELX)

Data Platform Architect *September 2020 - August 2022*

Led architecture development for the Entellect data platform, supporting enterprise data integration initiatives.

- Provided architectural guidance for platform development and integration
- Facilitated collaboration between product and engineering teams to align technical capabilities with product strategy
- Supported pharmaceutical company data-athon, enabling novel drug target discovery through platform integration

Honeybee Solutions (Synchronoss Technologies Inc.)

R&D Lead, Architecture Team / Senior Platform Architect *August 2017 - June 2020*

Developed a rules-based product recommendation service and integrated a schemaless product catalog with a recommendation engine.

- Developed a rules-based solution for a product recommendation service using Datalog derivatives.
- Integrated a domain-specific layer for a schemaless product catalog with a recommendation engine using JSON-LD and RDF.

Janrain Inc.

Independent Contractor *April 2017 - July 2017*

Built a rule-based system with Clojure, ClojureScript, and Datomic for runtime business logic modifications without core code changes.

- Built a rule-based system with Clojure, ClojureScript, and Datomic, enabling runtime business logic modifications without core code changes.

Eckoh plc

Head of Research & Development *September 2013 - November 2016*

Managed an R&D team, exploring new technologies and advising on best practices, focusing on data-driven architectures and technology advancements.

- Managed a small R&D team, exploring new technologies and advising on best practices.
- Focused on data-driven architectures and industry-academic technology advancements.

BraveNewTalent

Senior Developer / Team Lead *January 2012 - August 2013*

Led a scrum team transitioning from a monolithic platform to discrete services, introducing technologies like ElasticSearch, Jenkins, and Vagrant.

- Led a scrum team, transitioning from a monolithic platform to discrete services.
- Introduced new technologies including ElasticSearch, Jenkins, and Vagrant.

Eckoh plc

Senior Developer *June 2005 - January 2012*

Developed and maintained the PHP/VoiceXML platform, implemented coding and security standards (PCI/DSS), and mentored development teams.

- Developed and maintained the PHP/VoiceXML platform.
- Implemented internal coding and security standards (PCI/DSS) and mentored development teams.

365 Corporation plc / Eckoh plc

Developer *October 1999 - June 2005*

Maintained and improved a C++ IVR platform, enhancing uptime and developing voice applications.

- Maintained and improved a C++ IVR platform, enhancing uptime and developing voice applications.

Total Perspective Tech

Founder & Lead Developer *June 2023 - Present*

Exploring the intersection of emerging and forgotten concepts in software through open-source projects and long-form writing. Recent threads include relational data systems, type-level programming, and agent-native development methodology.

- Founded Total Perspective Tech as the home for open-source work and applied research.
- Seeded the RelationalFabric collective (Canon, Howard, Suss) — foundational TypeScript libraries for truly relational, data-centric systems.
- Built Agent Brain Trust — a reusable system of expert collectives that turns AI assistants into a room with friction, shipping as Cursor/Claude Code plugins and an MCP server.
- Published a sustained body of long-form writing on Level Up Coding spanning platform thinking, knowledge engineering, type-level programming, and AI-assisted development.

Technical Skills

Programming/Domain Specific Languages: C/C++, Clojure, ClojureScript, Datalog, Java, JavaScript (ES6), Node.js, Perl, PHP, Python, SPARQL, SQL, TypeScript.

Web and Markup Technologies: CCXML, CSS, GraphQL, HTML, JSX, Next.js, Pinia, Pug, React, Redux, Schema.org, Svelte 4, SVG, TailwindCSS, VoiceXML, Vue/Nuxt, Vuetify, WebRTC, WebWorkers.

Databases and Data Management: Datomic, ElasticSearch, FlureeDB, MongoDB, MySQL/MariaDB, Postgres, Redis.

Development and Deployment Tools: AWS, AWS API Gateway, AWS Cognito, AWS DynamoDB, AWS ECR, AWS ECS, AWS EKS, AWS Fargate, AWS Lambda, AWS S3, AWS SNS, AWS SQS, Chromatic, Docker, ESLint, GitHub Actions, Histoire, Kafka, Langium, N8N, NATS, RabbitMQ, SST (Ion), Storybook, Terraform, ZooKeeper.

Artificial Intelligence and Machine Learning: Agent Brain Trust, Agent-native architecture, Agentic AI/LLMs, AI planning (symbolic), ChatGPT, Claude.ai, Claude Code, Cursor.sh, Custom GPTs, DeepSeek, Dialectic prompting, Google Colab, Google Gemini, MCP (Model Context Protocol), NotebookLM, oLLAMA, OpenAI APIs, Perplexity, production-rule systems, Prompt Engineering, PyTorch, statistical methods, synthetic data generation, Tensorflow, Vector Symbolic Architectures, WebLLM.

Methodologies: Atomic Design, BDD, Design Systems, Domain-Driven Design, Kanban, Lean, SAFe, Scaled Agile, Scrum, TDD.

Architectures: Bloom / CALM, C4, Curry–Howard correspondence, data-driven architectures, Distributed architectures, event/streaming architectures, Lazy typing, micro-service architectures, Monotonic reasoning, propagator networks, Relational programming, serverless architectures, Stratified reasoning, Type-level programming, Value semantics.

Compliance: GDPR, PCI DSS.

Additional Tools: Figma, Notion, PostHog.

Knowledge Engineering: JSON-LD, OWL, RDF, Schema.org, Semantic Web.

Education

BSc in Physics Imperial College London

Projects

Linear Project Manager

Status: Completed

Type: Proprietary

Site: https://chatgpt.com/g/g-BvCW2ngMb-linear-project-manager

Description:

Custom GPT designed to assist with Linear platform project management via GraphQL API. Provides guidance for query construction, schema validation, and API usage examples.

- Manages projects using the Linear platform's GraphQL API
- Constructs and executes queries and mutations for issues, projects, teams, and users

Groove Dojo

Status: Work in Progress

Type: Proprietary

Description:

AI-enhanced music playlist management system with Spotify integration. Focuses on personalized playlist creation based on user preferences and context.

- Developing Design System using Svelte, shadcn-svelte, and TailwindCSS
- Planning AI integration for music recommendations
- Implementing collaborative playlist features
- Developing dynamic playlist generation
- Integrating with Spotify API for user authentication and playlist management
- Using linked data for data representation and FAIR data storage

Nuxt Gravity

Status: Work in Progress

Type: Proprietary

Site: https://levelup.gitconnected.com/from-atomic-design-to-relativistic-interfaces-0751b0b46832

Description:

Nuxt module implementing Mouldable UI components, based on an extension of Atomic Design principles for flexible interface development.

- Developed Relativistic Interfaces concept extending Atomic Design a
- Created Nuxt module for Mouldable UI components
- Currently implemented in production design system
- Open source version being implemented from lessons learned in a proprietary project

Pondermatic

Status: Active

Type: Open Source

Repository: https://github.com/totalperspective/pondermatic

Description:

Effect system based rules engine for Clojure/ClojureScript, focusing on functional programming principles.

- Developed an opinionated rules engine for Clojure/ClojureScript
- Integrated with Asami for data storage and querying
- Leans on the O'doyle rules-engine for RETE based rules execution
- Added support for JavaScript environments
- Published to Clojars and npm repositories (JS support out-of-the-box)

Oolon

Status: Superseded

Type: Open Source

Repository: https://github.com/totalperspective/oolon

Description:

Bloom implementation for Clojure/ClojureScript using Datomic datalog. Project concepts now incorporated into Pondermatic.

- Created declarative language for state management
- Implemented modules for distributed system state handling
- Used DataScript for internal state management
- Developed data-oriented DSL for Java/JavaScript integration

Bartfast

Status: Superseded

Type: Open Source

Repository: https://github.com/totalperspective/bartfast

Description:

Design language development toolkit focusing on modularity and reusability in digital design. Concepts now part of Relativistic Interfaces.

- Developed modular design component system
- Created declarative approach to design language definition
- Implemented design token and principle definition system
- Added language server for development support
- Superseeded by work on Relativistic UI

Agent Brain Trust

Status: Active

Type: Open Source

Repository: https://github.com/bahulneel/agent-brain-trust

Description:

A reusable system of expert collectives ('trusts') that turns AI assistants into a room with friction — real named figures in plausible settings, a moderator who refuses to skip steps, and a synthesis that names what was traded away. Ships as Cursor and Claude Code plugins, per-skill zips, and a published MCP server (@bahulneel/brain-trust-mcp).

- Composable Agent Skills following the agentskills.io specification: expert-opinion, draft-experts, and ten bt-* workshops and editorials (software systems, codebase tactics, prompt engineering, design patterns, product strategy, organisation design, frontend UX, visual communication, technical writing, science explanation)
- Bounded guest drafting via an MCP-served persona taxonomy of ~80 named figures — closes off

invented authority and lets one room draft a witness from another's domain

- Shared protocol fragments (readings, synthesis, persona fidelity) reused across rooms; a new room is one YAML stanza and a paragraph of framing
- Monorepo of brain-trust-core, brain-trust-db, brain-trust-mcp (published as @bahulneel/brain-trust-mcp on npm) and repo-tooling, with releases producing Cursor and Claude Code plugin zips and standalone RPL.md / LRPL.md prompt artifacts
- Spawned RPL (Relational Prompt Language) as a separate, reusable artifact — see the RPL project entry

RPL — Relational Prompt Language

Status: Active

Type: Open Source

Repository: https://github.com/bahulneel/agent-brain-trust/tree/main/docs/rpl

Description:

A Markdown-embedded reasoning framework with associated logic for structuring how an LLM reasons about a user's request. Instead of relying on highly tuned prose and guessing what the model inferred, RPL lets you declare explicit relations, goals, and constraints so reasoning state becomes inspectable, queryable, and continuable across context boundaries. Spawned out of Agent Brain Trust and published as a standalone artifact.

- Stratified relational reasoning — each conversation turn is a stratum, the trace is the continuity mechanism between strata and across sessions
- Grounded in Bloom and the CALM theorem (Hellerstein; Ameloot, Neven & Van den Bussche): the monotonic core needs no coordination, with non-monotonic operations made explicit as agent-judgment points
- Base RPL (eager) and LRPL (lazy extension with memos, satisfactory quiescence, and a small stdlib including $json, $write, $index)
- Prose-first authoring: Markdown is materialised into relations on encounter; relation heads attach to headings only where prose drifts
- Distributed as published RPL.md / LRPL.md artifacts inside the Agent Brain Trust releases, with a normative base specification and tutorials

Relational Fabric

Status: Active

Type: Open Source

Site: https://github.com/RelationalFabric

Description:

Foundational TypeScript libraries and published research for building truly relational, data-driven systems — providing the ontological foundations and construction primitives for systems where relationships carry semantic meaning, different representations interoperate, and code can reason about its data relationally.

- Filament — meta-level primitives for building domain-specific abstractions
- Weft — navigation and query construction primitives
- Warp — primitives for working with data at rest
- Shuttle — flow and coordination construction primitives
- Published whitepapers: Relational Causal State Transition System (RaCSTS) — implemented by Suss; and Distributed Context-Sensitive Graph Store (DCSGS) — a peer-to-peer, semantically rich graph store combining an immutable Semantic Log with a Native Projection, audience-based stratification, and ownership-based conflict resolution

Canon (@relational-fabric/canon)

Status: Active

Type: Open Source

Repository: https://github.com/RelationalFabric/canon

Description:

Universal type primitives and axiomatic systems for TypeScript. Implements the Lazy Typing pattern: write type-safe code against semantic concepts (axioms) while deferring shape-specific implementations (canons) to runtime configuration, exposed through universal APIs.

- Axioms / Canons / Universal APIs — semantic concepts decoupled from data shape
- Zero-runtime-cost compile-time type assertions (Expect<A, B>, IsTrue, IsFalse)
- Bundled tsconfig, ESLint config and a CLI scaffold (`npx canon init`)
- Transparent ADRs and a public technology radar

Howard

Status: Active

Type: Open Source

Repository: https://github.com/RelationalFabric/howard

Description:

A self-contained logical primitive for building truly relational data systems — a computable truth engine that turns ad-hoc boolean validation into first-class, composable propositions, in the spirit of the Curry–Howard correspondence.

- Value semantics and verifiable equality for arbitrary data
- Claims backed by pure, deterministic predicates
- Compositional propositions mirroring domain relationships
- Decouples what is claimed from how it is proved, for a flexible and extensible architecture

Suss

Status: Active

Type: Open Source

Repository: https://github.com/RelationalFabric/suss

Description:

Reference TypeScript implementation of RaCSTS (Relational and Causal State Transition System) — serialisable propagator networks that can be paused, inspected, versioned and transmitted across boundaries as first-class values.

- Bidirectional constraints with automatic reconciliation
- Leader-free consensus via Sync; causal ordering via a Hybrid Epoch Clock
- Six standard higher-order relations (mark, linear, map, constrain, reduce, join)
- Full pack/unpack of networks as values for portability and replay